

SHOW ME
YOUR AGENTS
HACKATHON

Introduction to the Hackathon

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Who are we?

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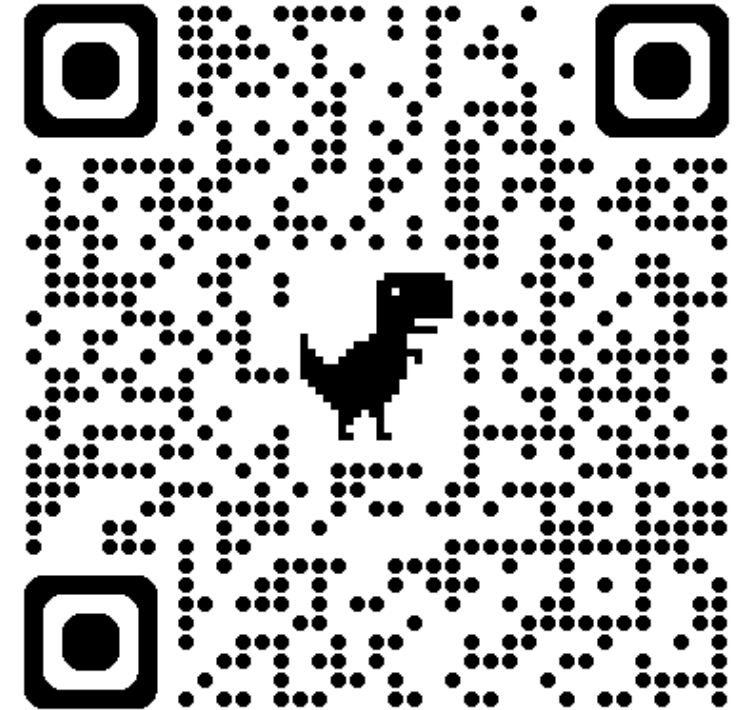
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We are from the Software Systems Practice

There are 16 of us, software practitioners who are lecturers and consultants.

We are passionate about sharing our knowledge and skills, from GenAI and Agentic Solutions to Software Engineering best practices, through Corporate/Executive classes and NUS Graduate Programmes.



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**What will you gain
from this hackathon?**

A Pathway to Building AI Agents

We believe your next team member could be an AI Agent!



Product Manager

Owns the roadmap. Decides what gets built and why, balancing user needs with business goals.



Architect

Designs the system blueprint — how components connect, which technologies to use, and how to scale.



Data Engineer

Builds the pipelines that collect, transform, and deliver data reliably to the right systems.



UX/UI Designer

Turns user research into intuitive interfaces — designs and validates experiences before code is written.



Developer

Writes production-grade code that is readable, tested, and maintainable by the whole team.



QA / Test Engineer

Owns quality throughout the lifecycle — not just at the end. The last line of defence before users.



Security Engineer

Embeds security at every phase — from threat modelling in design to penetration testing before release.



Code Reviewer

Reviews every code change before it merges — enforcing standards and catching defects early.



DevOps Engineer

Automates build, test, and deploy pipelines. Manages infrastructure and owns incident response.

Next Gen Software Engineering Team

We are reimagining Software Engineering Skills with Agentic AI

No Code Creator

Target audience: non-Tech, startup, entrepreneurs
Needs: quick delivery of products

Early adopters

(both solution architect and developer)

Target audience: already using Agentic AI tools in their day-to-day work
Needs: quick tips on what can be done and how

Experienced SW professionals (Tech leads/mgrs/snr architects)

Target audience: experienced in building and maintaining software systems, familiar with agile practices
Needs: how to introduce agentic AI tools and yet remain resilience and enforce governance

1

Building no code Agent for Workflow Redesign

3

Building autonomous AI agents

5

Build a Secure and Reliable Agentic AI System

2

AI-Augmented SDLC – From Copilots to Agentic Pipelines

4

Architecting AI solutions

6

Integrating and Deploying AI Solutions

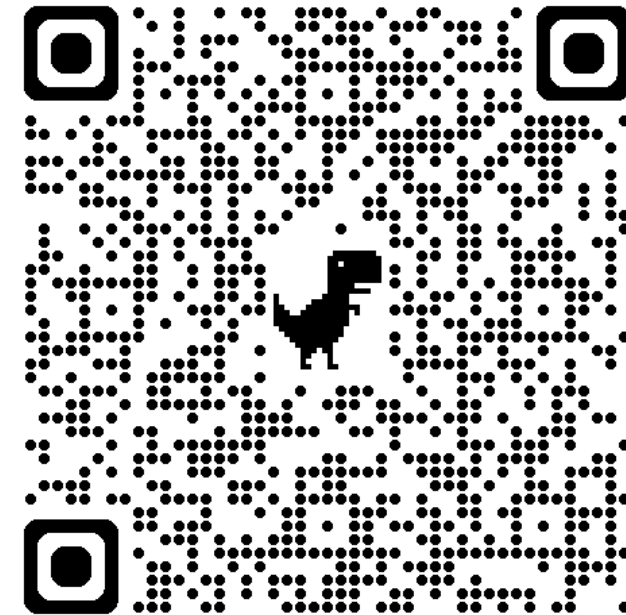


This is a suggested pathway, courses can be taken in different order

Fundamental (Complete 2 Graduate Certificates)		Specialist (Complete 2 of 4 Graduate Certificates)			
Designing Modern Software Systems	Architecting Scalable Systems	Architecting AI Systems	Designing and Managing Products and Platforms	Engineering Big Data	Securing Ubiquitous Systems
Essential Practices for Agile Teams	Architecting Software Solutions	Explainable and Responsible Artificial Intelligence	Service Design	Information Architecture for Data-driven Insights	ISC2 CISSP CBK Training Seminar
Software Analysis & Design	Platform Engineering	AI and Cybersecurity	Managing Digital Products	Big Data Engineering for Analytics	Secure Software Development Lifecycle for Agile
Software Design Patterns	Cloud Native Solution Design	Architecting Agentic AI Solutions	Digital Product Strategy	Architecting Systems for Real-Time Data Processing	Design Secure Mobile Architecture
DevSecOps Engineering and Automation		Deploying and Operating AI Solutions	Architecting Platforms as a Business		Platform Security
Graduate Certificate in Designing Modern Software Systems	Graduate Certificate in Architecting Scalable Systems	Graduate Certificate in Architecting AI Systems	Graduate Certificate in Designing and Managing Products and Platforms	Graduate Certificate in Engineering Big Data	Graduate Certificate in Securing Ubiquitous Systems

Master of Technology in Software Engineering

(Through train or stackable)



Watch this space for something exciting!

**Graduate
Certificate in
Digital Solutions
Development -
Design**

Software Analysis
and Design, Software
Engineering (UX,
requirement
analysis, software
design, agile,
SCRUM)

**Graduate Certificate in Digital Solutions
Development – Agile Software Delivery**

DevOps Practices, DevSecOps consideration and MLOps

**Graduate Certificate in Digital Solutions
Development – Web Application**

Web Application Development: JAVA, RESTFUL server
side, web design, HTML and CSS

**Graduate Certificate in Digital Solutions
Development – Mobile Application**

Developing Mobile Solutions with AI & Agents

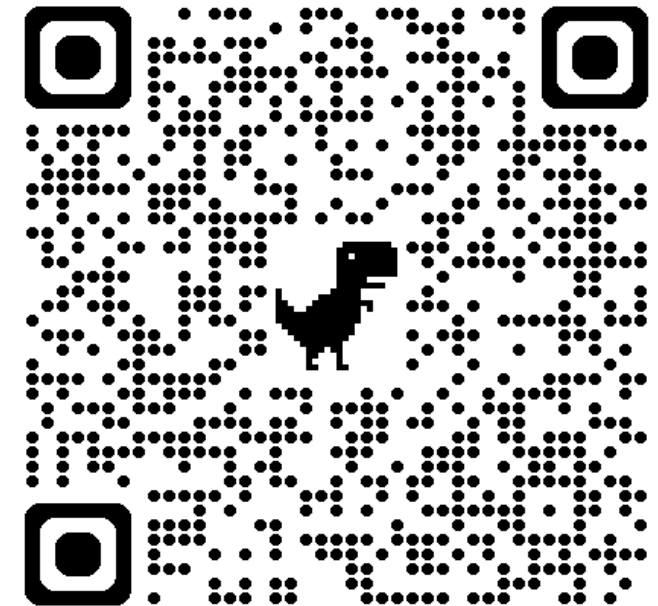
**Graduate Certificate in Digital Solutions
Development – Machine Learning Application**

Machine Learning Application Development: Machine
learning application design and development using
Python

Graduate Diploma in Systems Analysis – Capstone and Internship

Capstone project delivery and 5-months internship: Gaining relevant professional
experience, project management skills, and building soft skills in actual work
environment.

Graduate Diploma in Systems Analysis



Updates on the way

GenAI & Agentic AI



Architecting AI Systems



Digital Solutions Development



Software Architecture



Architecting Scalable Systems



Product Management



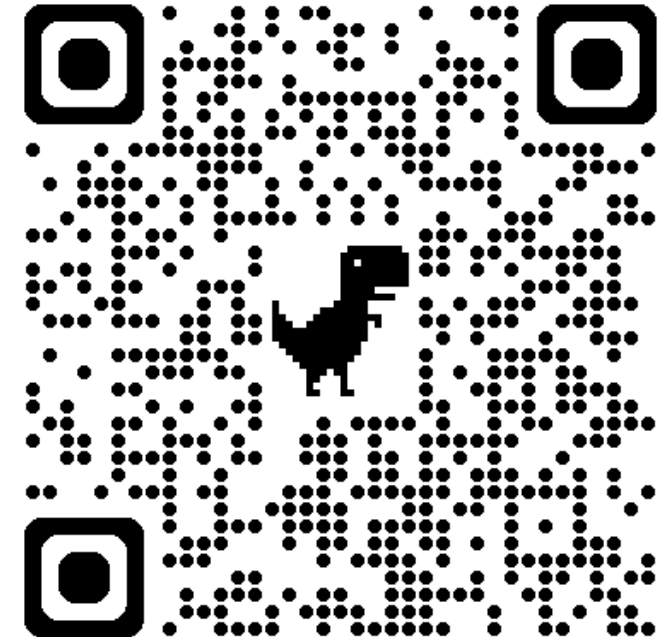
Engineering Big Data



Securing Ubiquitous Systems



Visit us to find out more



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Hackathon Briefing

Two Categories, One Mission

SME Category

Open to teams representing Small and Medium Enterprises. Suitable for SME owners, employees or official representatives building an AI agent solution for their own business challenges.

- Teams must represent an SME
- Annual sales turnover should not exceed S\$100 million
- Proof of SME affiliation may be required at registration
- Each team must have 2 to 4 members
- At least one member should have technical skills

Public Category

Open to individuals not currently employed by or representing an SME. Including students, researchers, startups, professionals and tech enthusiasts.

- Each team must consist of exactly 4 members
- Teams work on official SME business problem statements shared during the hackathon kickoff
- Additional recognition or scoring consideration for effectively addressing real SME problems

Prizes

Prizes (Per Category)

- 1st Place: USD 3,000 AWS cloud credits
- 2nd Place: USD 2,000 AWS cloud credits
- 3rd Place: USD 1,000 AWS cloud credits

Special Awards

- SME Ready Award (for SME Category): SGD 1,000 cash
- Best Agent Use Award: SGD 1,000 cash
- Social Impact Award: SGD 1,000 cash

Judging Rubric

1. Goal & Scope Definition

- e.g., business value, clear purpose

2. Architecture & Reasoning Loop

- e.g., appropriate planning pattern, explicit state/memory management

3. Tool Use & Integration

- e.g., purpose-fit tools, well-typed schemas

4. Autonomy & Human-in-the-Loop

- e.g., risk-calibrated autonomy, escalation checkpoints, etc.

5. Safety, Security & Guardrails

- e.g., prompt-injection resistance, least-privilege access

6. Observability & Evaluation

- e.g., logging/tracing of decisions, golden-path + adversarial eval cases, etc.

7. Platform & Tooling Usage

- e.g., idiomatic framework use, clean multi-agent orchestration, etc.

How to Participate

- **Step 1:** Join the Slack workspace
 - Check Slack frequently for updates and announcements
 - All announcements, submissions and support happen on Slack
- **Step 2:** Submit your business proposals (if you have not done so)
- **Step 3:** Attend the AWS Kiro training by AWS
- **Step 4:** Take note of the submission date
- **Step 5:** Upload the submission via the Slack channel in the required format

Key Dates – Build Phase and Finale

Watch Slack for announcement

Shortlisting Round

Submission deadline: **28 September 2026, 9:00am**

Finale

Be ready for a face-to-face demo on **10 October 2026, 8:30am**

Submission Format – via Slack Channel

- Post each round's submission to the **designated Slack channel** in this format:
 - Team Code
 - Project Name
 - GitHub Repo (URL)
 - YouTube URL, or URL to download the video in MP4 format (duration: 30mins)
- Write up document in PDF
- Deployment evidence/URL

Note that incomplete submissions may be rejected, so please follow the format accordingly.

AWS Resource & Kiro Credits

Shortlisting Round

7 – 28 September 2026

- AWS credits: USD 100
- Kiro credits: USD 20 per participant

Finale

6 – 10 October 2026

5 public teams + 5 SME teams

- AWS tokens: 100
- Kiro credits: USD 20 per participants

The Kiro credits and Zoom link emails will be sent out after today's briefing.

AWS Resource Usage Model

- AWS resources are provided for deployment and LLM prompts
- Participants are advised to develop their work locally using their own AI client tools (e.g., Kiro, Claude Code, Cursor, Copilot, etc.)
- Allowed AWS usage:
 - AWS Lightsail
 - JSON API calls to access AWS Bedrock – Claude Sonnet 4.5 model
 - JSON format will be provided via Slack
 - API Usage plan will be shared over Slack
 - API key for each team will be sent via email

If you exceed this AWS usage limit, your account may be paused, which could affect your standing in the competition.

AWS Kiro Training by AWS

- AWS will conduct a Kiro online training session for all participants
- Learn how to build your agent with Kiro
 - 6 September 2026, 10am to 12pm
 - Zoom link will be announced on Slack
- Workshops and support activities will be announced on Slack

Tools Support – via Slack

- Stuck with your agent tooling? Help is provided via Slack:
 - **OpenClaw**
 - **Hermes Agent on AWS**
 - **NanoClaw**
- Post your question on the Slack workspace for support

Build Your Own Custom AI Agent

- Participants are allowed to code their own AI Agent
- Any framework
- Any programming language
- You are free to choose the tools that best fit your project



Slack Invitation



Proposal form

Please join Slack and submit your proposal before 28 September 2026 - 9am

Our Slack Support Channel

Slack Channels	Description
#tech-support	Technical help channel for AWS-related issues and other technical questions participants run into.
#solutioning	Where teams discuss approach, architecture, or brainstorm solutions to the problem statement.
#admin-related	General help channel where participants will ask organisers or mentors questions.
#submission	Where teams submit or discuss their deliverables.
#final-submission	Submissions for the final round or finalists.

Check Slack Frequently!

All announcements, key dates, Q&A and support will be on the Slack workspace.



The background image is a complex digital dashboard titled "AI AGENT SUPERVISOR: STATUS OVERVIEW". It features several data visualization components: a "NEURAL NETWORK MODEL 07" diagram with nodes and connections; a "TASK ALLOCATION" bar chart showing data for months from JAN to DEC; a "SYSTEM ALERTS" list with timestamps and messages; a "LANGUAGE PROCESSING" section with text analysis; a "PERFORMANCE METRICS" waveform graph; and an "AGENT NETWORK" diagram. The central part of the dashboard displays the status of various agents: "AGENT 01: PROBLEM SOLVER" (ACTIVE), "AGENT 02: RESOURCE MONITOR" (ACTIVE), "AGENT 03: PROBLEM SOLVER" (PENDING), and "AGENT 04: SPOKESMAN MONITORING" (PENDING). A large white oval with the text "Thank You" is superimposed over the center of the dashboard. At the bottom, silhouettes of seven people are shown, some holding tablets, as if they are interacting with the system.

Thank You

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